Educational prototype. Not a medical device. Not clinically validated. All demo cases are synthetic.

KidCheck

Plain words in, WHO-backed decisions out.

Live demo kidcheck-three.vercel.app
Code github.com/a3gissss/kidcheck
DSH Hacks V2 · AI x Healthcare · Abdulhafeez Ibrahim

The problem

In rural communities, including much of Nigeria, the first person to see a sick child is often a community health worker with a few weeks of training and a paper chart. Under pressure, danger signs such as chest indrawing or not being able to drink get missed, and the child reaches a clinic too late.

The solution

KidCheck is a web app that helps the health worker check every danger sign in a child aged 2 months to 5 years and decide: **refer urgently**, treat at home, or no illness sign found. AI handles only language. Fixed, tested rules from the WHO/UNICEF chart booklet make the decision and show why.

How it works

1. Describe
The CHW writes what the carer said, in plain words. No names.

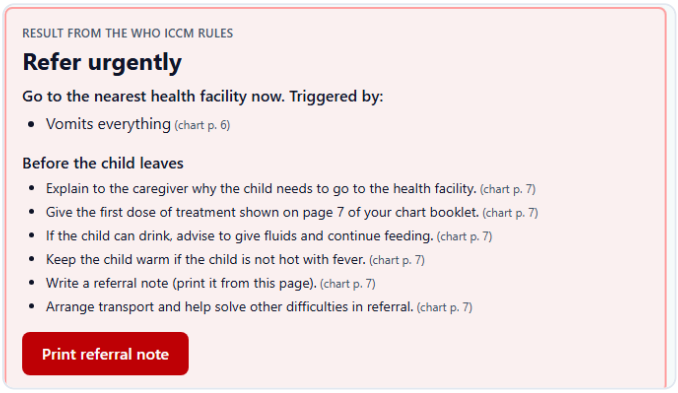
2. AI fills
Gemini turns the text into checklist answers, each marked AI. Unmentioned signs stay "Not asked".

3. CHW checks
Confirms or corrects every answer; adds breath count, chest indrawing, MUAC.

4. Rules decide
The WHO iCCM rule engine gives the result, the triggering sign and the chart page.

Key features (all working)

- AI checklist filling with an **AI** mark on every machine answer
- Rule engine for the WHO/UNICEF danger signs, fast breathing, diarrhoea, fever and MUAC, citing booklet pages
- "Not asked" is never treated as "No": unchecked danger signs give *Needs more answers*
- Full-screen urgent alert, pre-referral steps, printable referral note
- Case log on the device; edit, filter, delete
- Manual checklist when the AI is unavailable, with identical rules
- Blocks phone numbers before sending to AI; works on phones; accessible



Synthetic case: "Boy, 18 months, coughing 3 days, breathing fast, vomits everything" → Refer urgently (chart p. 6).

Technologies

React 19 · Vite · Tailwind CSS v4 · React Router · Google Gemini API (`@google/genai`) · Vercel serverless function and hosting · localStorage · Vitest, Testing Library, axe-core · MIT licence

Target users

Primary	Community health workers and volunteers in low-resource settings
Also	Nursing and public health students practising assessments
Setting	Phone or laptop at the point of care; printed note for the facility

AI and technical approach

AI	Gemini (`gemini-3.5-flash-lite`, free tier) with a JSON schema built from the checklist; output re-checked on the server
Decision	Plain JavaScript rule engine, no AI, 56 unit tests
Privacy	Key on the server only; cases stored in the browser; no names
Quality	168 automated tests incl. full-journey UI tests; accessibility scan clean

106/106 stated facts extracted
0 danger signs missed
168 automated tests

AI figures: 20 synthetic, developer-labelled cases incl. Nigerian Pidgin. Not real-world accuracy.

Expected impact

Fewer missed danger signs, because every sign is asked about and an unanswered one blocks the decision. Faster referral, because the result is immediate and explains itself. A record of each case where there was none. Low cost: free tools, a phone browser, and no patient data leaving the device.

Rules: WHO/UNICEF, *Caring for the sick child in the community*, CHW chart booklet (2014). Not reviewed by a clinician; doses are never shown.

Limits: ages 2–59 months; no HIV/TB questions; malaria area assumed.